Xi (Robby) Qiu

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Work Experience

Senior Machine Learning Engineer – PayPal *San Jose, CA* | **Jan 2025 – Present**

- Built an AI agent (Python, Gemini, Next.js) for financial risk analysis. Orchestrated agentic workflows for risk signal identification, data aggregation, and report drafting. Developed a copilot interface that allows investigators to review and correct AI-generated analyses at each stage. Improved report accuracy by 50% through context engineering and tool-call optimization.
- Built an end-to-end PySpark pipeline on GCP Dataproc to extract, summarize, and cluster risk factors from investigation reports using Gemini. Productionized the research prototype and cut runtime by 10x through batched LLM inference and parallel execution.
- Optimized LLM inference to support 700K+ daily calls in production. Benchmarked token throughput across Gemini and self-hosted Llama models using synthetically scaled datasets. Achieved 20% throughput gain via speculative decoding and deployed optimized vLLM serving stack on cloud GPU instances.
- Led the team's first LLM monitoring effort by deploying Langfuse-based observability across 10+ production solutions, providing end-to-end tracing of model outputs, cost, and token throughput at each pipeline stage.

Software Engineer (Platform Infrastructure) – PayPal *San Jose, CA* | **Jul 2023 – Jan 2025**

- Built and maintained 3 core infrastructure components powering PayPal's rewards platform (300M+ users): reward program management, real-time transaction processing, and batch payouts.
- Rebuilt the team's test infrastructure by enabling parallel execution and migrating test cases from Excel to Java. Reduced test execution time by 18x and improved traceability of test case changes.
- Improved production stability for the rewards platform's distributed services by triaging cross-team issues, debugging data pipeline failures, and resolving production incidents.

Education

University of Pennsylvania Aug 2021 – May 2023

- *Master of Computer and Information Technology* – GPA: 4.0

University of Oxford Oct 2015 – Jun 2019

- *BA in Philosophy, Politics and Economics* – First Class Honours

Projects

Agentic Capability Evaluation for Claude Opus 4.8 | *Claude, Hawk, InspectAI* | Link

- Evaluated Claude Opus 4.8 on 150+ coding, research engineering, and cybersecurity tasks (HCAST, RE-Bench) using Hawk and InspectAI. Estimated the model's capability ceiling by modeling success rates against task difficulty.
- Audited task transcripts to analyze performance deltas against the predecessor model, distinguishing genuine capability improvements from spurious results caused by reward hacking, safety refusals, instruction-following failures, and scorer bugs.

Alignment Evaluation for Frontier Models | *GPT, Claude, InspectAI* | Link

- Generated a dataset of 300 questions using Claude and GPT to measure LLMs' self-preservation behavior. Scored and filtered questions with LLM-as-a-judge to ensure that the dataset is valid, diverse, and unbiased.
- Evaluated GPT-4o mini's self-preservation tendency using InspectAI. Analyzed its moral decision-making capability across different prompting strategies including zero-shot, chain-of-thought (CoT), and CoT with self-critique.

RLHF for GPT-2 | *PyTorch, TransformerLens* | Link

- Implemented end-to-end PPO and GRPO algorithms for training GPT-2 with RLHF, including value head modeling, rollout collection, advantage estimation, and KL-regularization.
- Fine-tuned GPT-2 with both full-parameter training and LoRA. Experimented with rule-based and sentiment-based reward functions to analyze how distinct reward signals steer model behavior.

Transformer Interpretability Research | *PyTorch, TransformerLens, CircuitVis* | Link

- Recreated the GPT-2 model architecture from scratch based on the original paper. Trained on the `TinyStories` dataset and experimented with various sampling methods including temperature, top-k, and top-p.
- Detected induction heads with attention pattern visualization and ablation in GPT-2 small. Reverse-engineered QK-, OV-, and K-composition circuits using factorized matrices of query, key, and value weights of the transformer heads.
- Implemented sparse autoencoders to interpret GPT-2 layers, identifying monosemantic features and analyzing attention-layer activations.

RAG App for Rock Climbing Trip Planning | *Python, LangChain, Vector DB, Scrapy* | Link

- Developed a RAG recommendation system for rock climbing trips. Scraped data into a vector database for retrieval.